20251212 Sec. 10.6

ex 1 p. 635

$$u_{xx} = u_t, \quad 0 < x < 30, \quad t > 0$$

BC: $u(0,t) = 20, \quad u(30,t) = 50, \quad t > 0$

IC: $u(x,0) = 60 - 2x, \quad 0 < x < 30$

Consider $v(x) = 20 + x, \quad 0 < x < 30$

b/c it is the steady state sol'n $\left(t \to \infty, \ \frac{\partial u}{\partial t} = 0\right)$

which satisfies $v_{xx} = 0$

$$v(0,t) = 20, \quad v(30,t) = 50$$

Let $w = u - v$, then

$$w_{xx} = w_t$$

$$\left.\begin{aligned} w(0,t) &= u(0,t) - v(0,t) = 20 - 20 = 0 \\ w(30,t) &= u(30,t) - v(30,t) = 50 - 50 = 0 \end{aligned}\right\} \text{Dirichlet BC}$$

$$w(x,0) = u(x,0) - v(x,0) = (6 - 2x) - (20 + x)$$

$$= 40 - 3x$$

The rest follows

P. 635 - 6 **Insulated Ends**

$$u_x(0,t) = u_x(L,t) = 0, \qquad t > 0$$

Set $u(x,t) = X(x)T(t)$, we get

$$X'(0) = X'(L) = 0$$

$$\begin{cases} X'' + \lambda X = 0 \\ T' + \alpha^2 \lambda T = 0 \end{cases}$$

① $\lambda < 0$, $X(x) = c_1 e^{+\sqrt{-\lambda}\,x} + c_2 e^{-\sqrt{-\lambda}\,x}$

$$\cdots \Rightarrow c_1 = c_2 = 0$$

② $\lambda = 0$, $X(x) = c_1 + c_2 x$

$$X'(0) = X'(L) \Rightarrow \cdots \Rightarrow X(x) = c_1$$

③ $\lambda > 0$, $X(x) = c_1 \cos(\sqrt{\lambda}\,x) + c_2 \sin(\sqrt{\lambda}\,x)$

$$X'(x) = -c_1\sqrt{\lambda}\sin(\sqrt{\lambda}\,x) + c_2\sqrt{\lambda}\cos(\sqrt{\lambda}\,x)$$

$$X'(0) = -c_1\sqrt{\lambda}\,\cancelto{0}{\sin(\sqrt{\lambda}\,0)} + c_2\sqrt{\lambda}\,\underbrace{\cos(\sqrt{\lambda}\,0)}_{=1} = 0$$

$$\Rightarrow c_2 = 0$$

$$X'(L) = -c_1\sqrt{\lambda}\sin(\sqrt{\lambda}\,L) = 0$$

$$\Rightarrow \sqrt{\lambda}\,L = n\pi, \quad n = 1, 2, \cdots$$

$$\Rightarrow \lambda = \left(\frac{n\pi}{L}\right)^2, \quad n = 1, 2, \cdots$$

$$\Rightarrow \quad X(x) = c_1 \cos\left(\frac{n\pi x}{L}\right)$$

when $\lambda = 0$, $T' = 0 \Rightarrow T(t) = c = \text{const.}$

take $u_0(x,t) = 1$

when $\lambda > 0$, $T' + \frac{\alpha^2 n^2 \pi^2}{L^2} T = 0$

$$\Rightarrow T(t) = T(0)\, e^{-\alpha^2 n^2 \pi^2 t / L^2}$$

So the general solution is given by

$$u(x,t) = \frac{a_0}{2} + \sum_{n \geq 1} a_n e^{-\alpha^2 n^2 \pi^2 t / L^2} \cos \frac{n\pi x}{L}$$

IC :

$$f(x) = u(x,0) = \frac{a_0}{2} + \sum_{n \geq 1} a_n \cos \frac{n\pi x}{L}$$

w/ $$a_n = \frac{2}{L} \int_0^L f(x) \cos \frac{n\pi x}{L}\, dx$$

$$\left(\text{indeed, } \frac{a_0}{2} = \frac{1}{L} \int_0^L f(x)\, dx \right)$$

<u>ex 2</u> (p. 638) $L = 25$,

$u(x,0) = x$, $\quad 0 < x < 25$

Neumann BC.

So the general sol'n is

$$u(x,t) = \frac{a_0}{2} + \sum_{n\geq 1} a_n e^{-n^2\pi^2\alpha^2 t/25^2} \cos\frac{n\pi x}{25}$$

$$\frac{a_0}{2} = \frac{1}{25}\int_0^{25} x\,dx = \frac{25}{2}$$

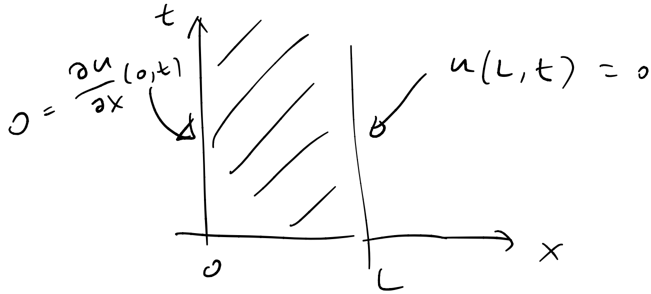
$n \geq 1$

$$a_n = \frac{2}{25}\int_0^{25} x\cos\frac{n\pi x}{25}\,dx$$

$$= \frac{50(\cos(n\pi)-1)}{(n\pi)^2} = \begin{cases} -\dfrac{100}{n^2\pi^2} & n \text{ odd} \\ 0 & n \text{ even} \end{cases}$$

Therefore

$$u(x,t) = \frac{25}{2} - \frac{100}{\pi^2}\sum_{n=1,3,5,\ldots} \frac{1}{n^2} e^{-n^2\pi^2\alpha^2 t/25^2}\cos\frac{n\pi x}{25}.$$



$\rightsquigarrow$ $X'(0) = 0$, $X(L) = 0$